

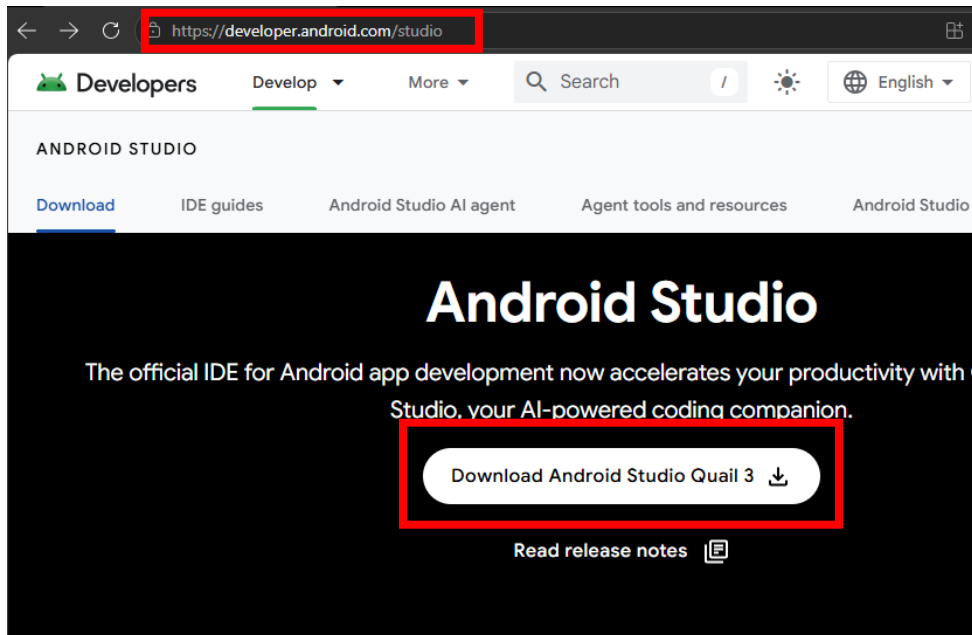
React Native Setup for Android

Base Requirements

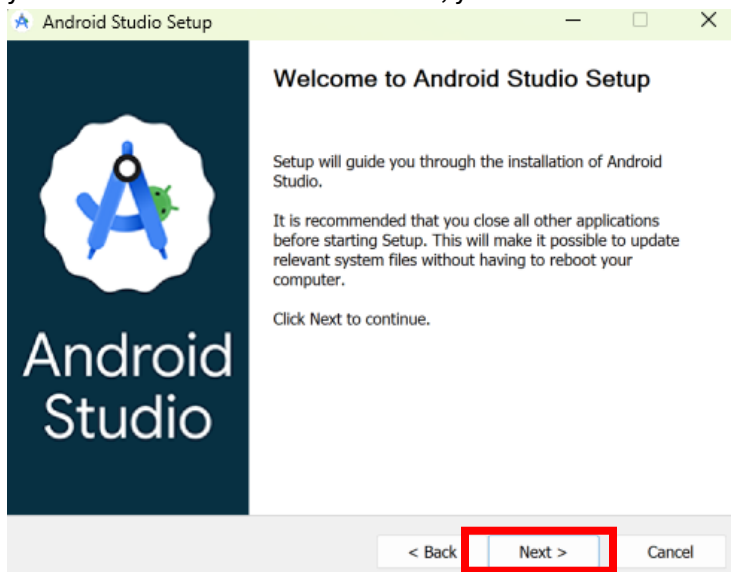
Ensure you have at a minimum Node.js 22 and npm 10 installed. You can download and install both from [Node.js — Download Node.js®](https://nodejs.org/en/download).

Download Android Studio

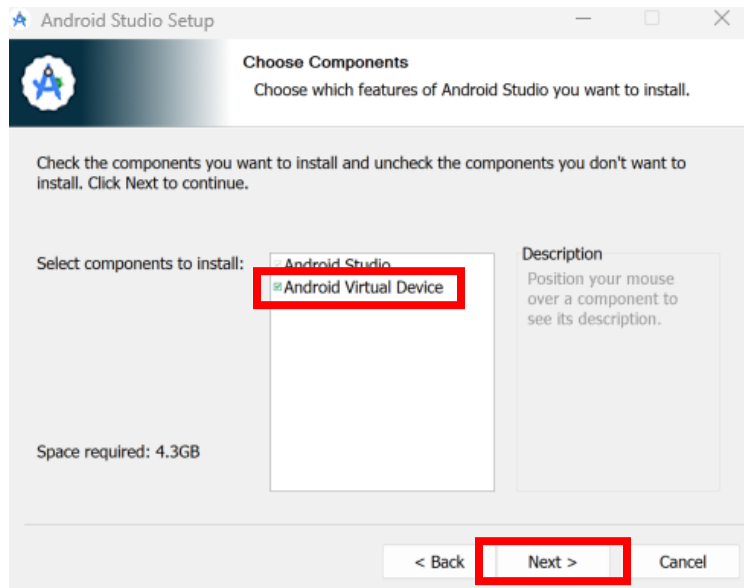
Open a browser and navigate to [Download Android Studio & App Tools - Android Developers](https://developer.android.com/studio), and click the Download Android Studio button (Android Studio Quail at the time of this writing). You will then have to accept the agreement and then download the installer.



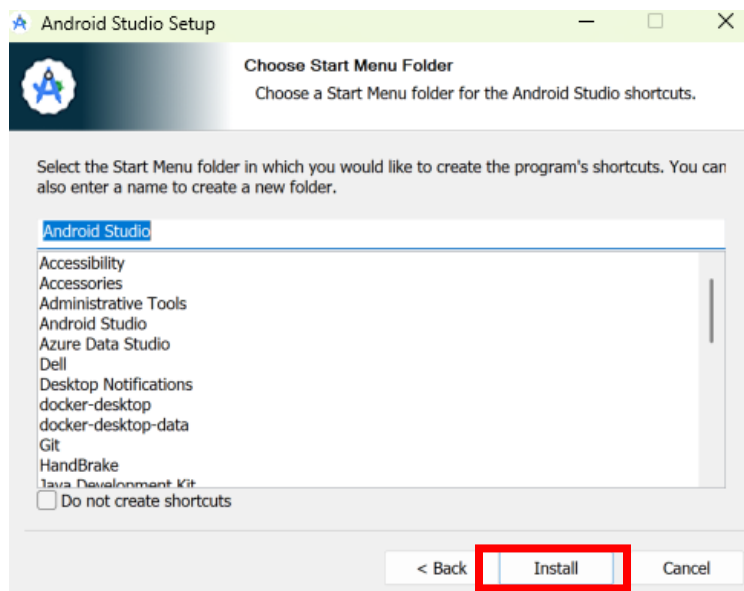
Once the installer has been downloaded (it may take some time), run it to install Android Studio. If you have an older version installed, you will be asked to remove it during the process.



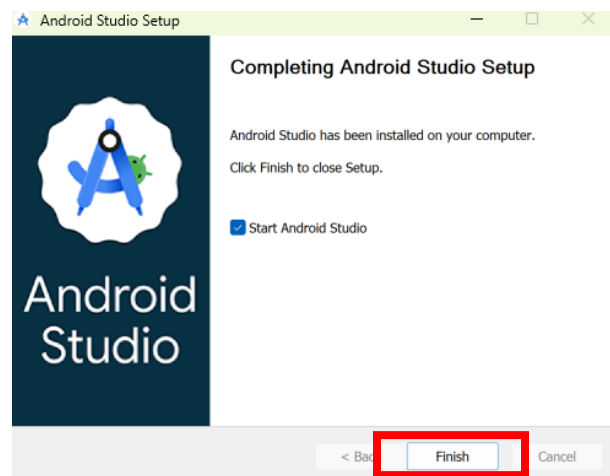
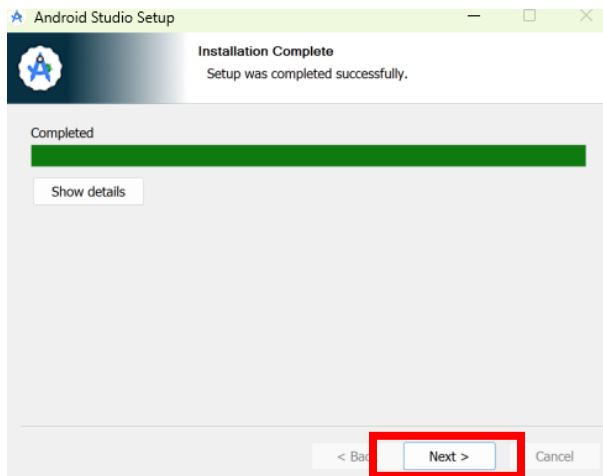
Hit the “Next” button and ensure that “Android Virtual Device” is checked on the next screen:



Click “Next” several more times until you reach the end of the install wizard and then click “Install”:

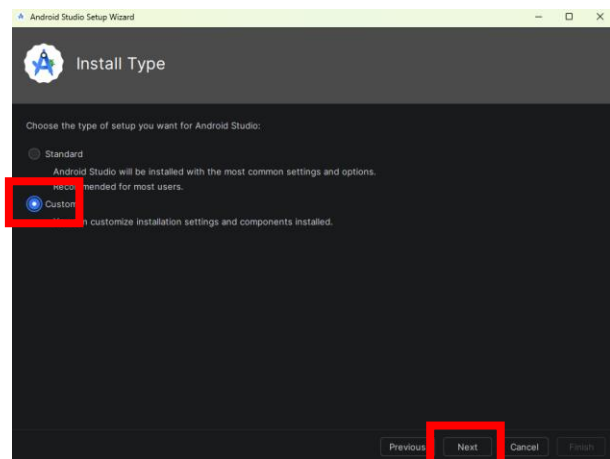
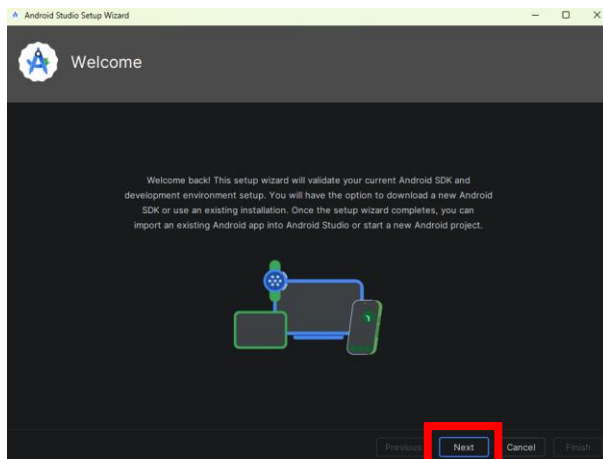


The install process may take some time. Once it's complete, click “Next” and then “Finish” to complete the process:

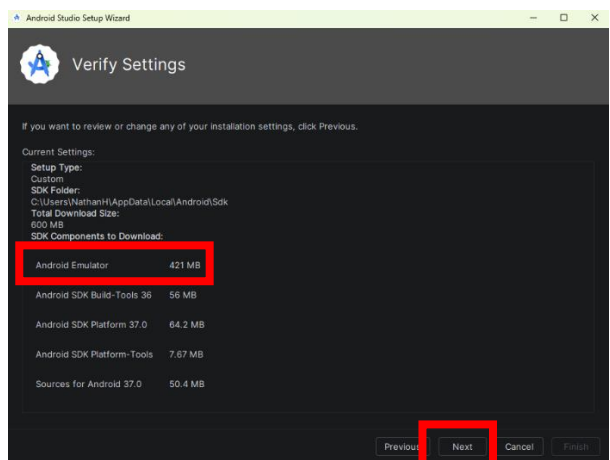
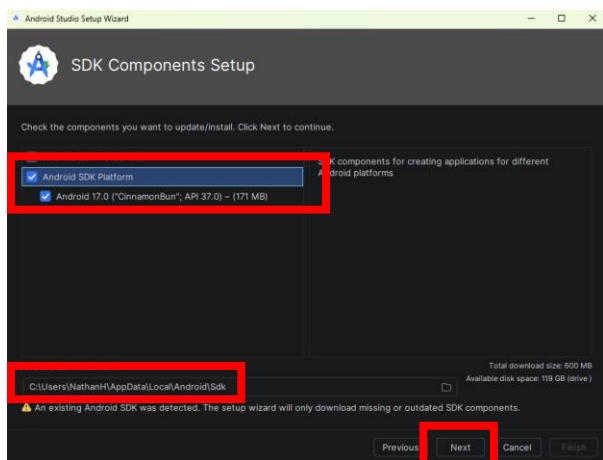


Configure Android Studio

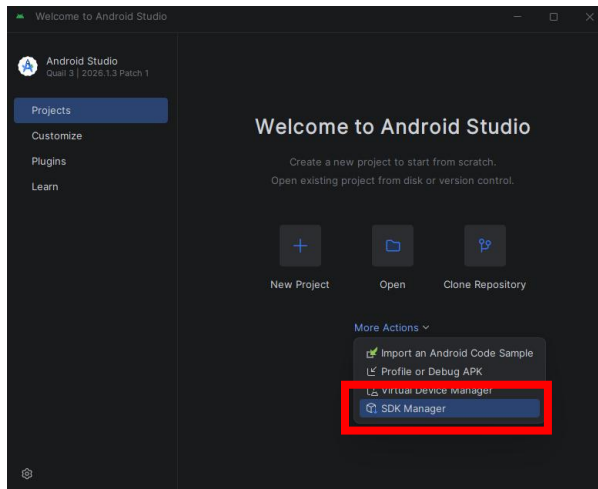
When you first open Android Studio, you will be presented with a wizard to help you set up the application. Click “Next” and on the second screen, choose “Custom” and then click “Next”.



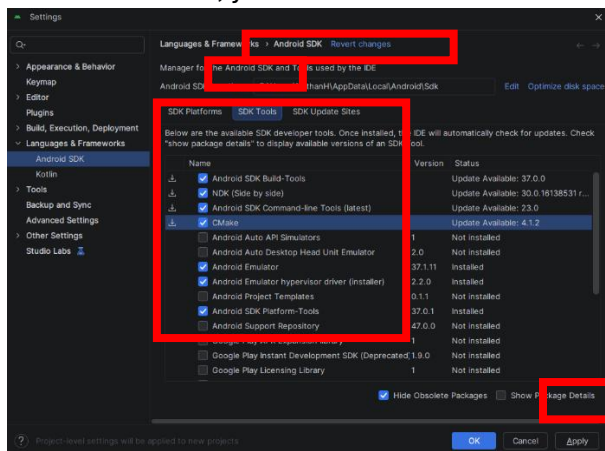
On the following screen, be sure to check “Android SDK Platform” and note the install directory location. You’ll be presented with a list of components; ensure that the emulator is listed and click “Next”. Once the downloading is complete, click “Finish” to complete the setup.



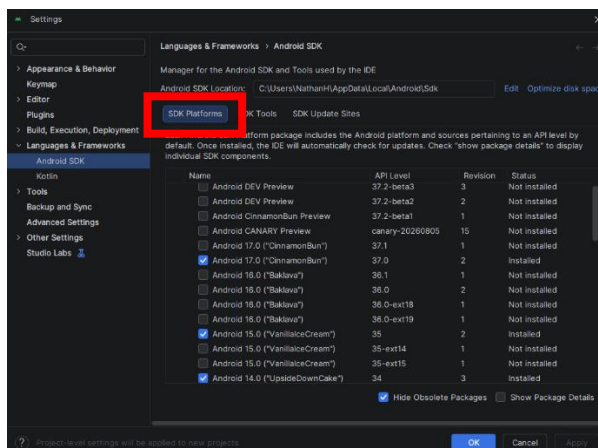
Once the initial components are installed, you should see the “Welcome to Android Studio” screen. Click “More Actions” and then click “SDK Manager”, you will need to verify you have the right components installed.



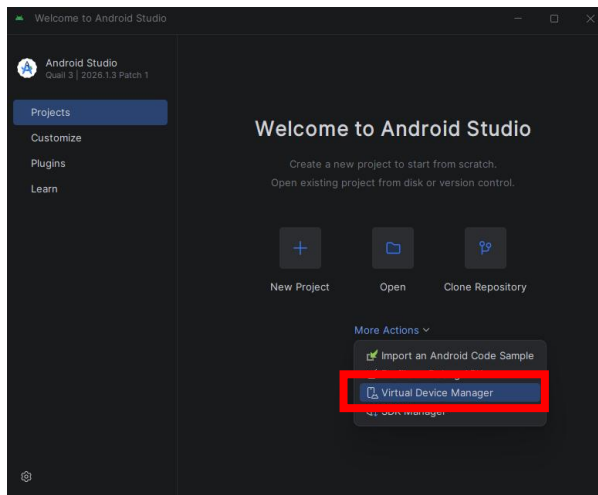
Navigate to the “SDK Tools” tab, and ensure the following options are selected. Also, take note of the SDK location, you will need it later.



If you scroll down here, you should also see “Intel x86 Emulator Accelerator (HAXM Installer)” and “SDK Patch Applier v4” checked. Click “Apply” and then “Ok” on the dialogue window to install the remaining components. Once the installation is complete, click “Finish” and you should be back on the Android SDK screen. Navigate to SDK Platforms to see the target platforms currently installed for you. If you need to target a specific Android version, this is where you can manage which versions are available to you.

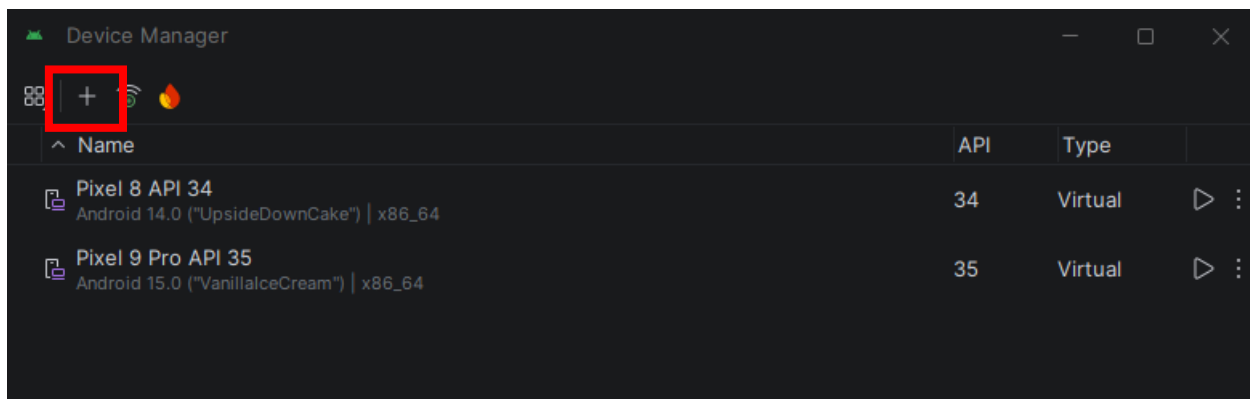


As you can see above, versions 14, 15, and 17 are installed on this machine. Your installed platforms may vary, but ensure that you have version 17 installed at a minimum. Click “Ok” and return the Welcome screen. From there, again click “More Actions” and select “Virtual Device Manager”.

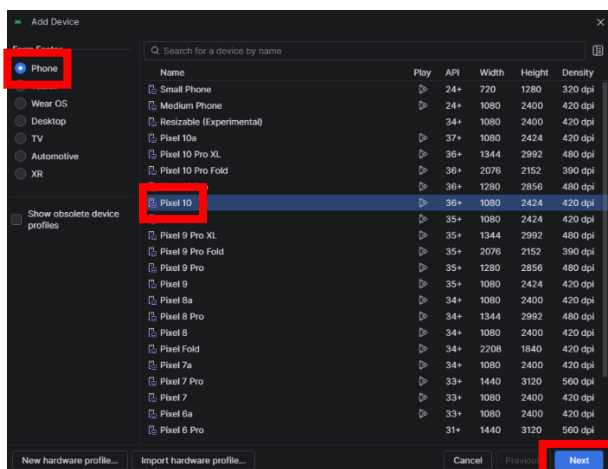


Create an Emulator

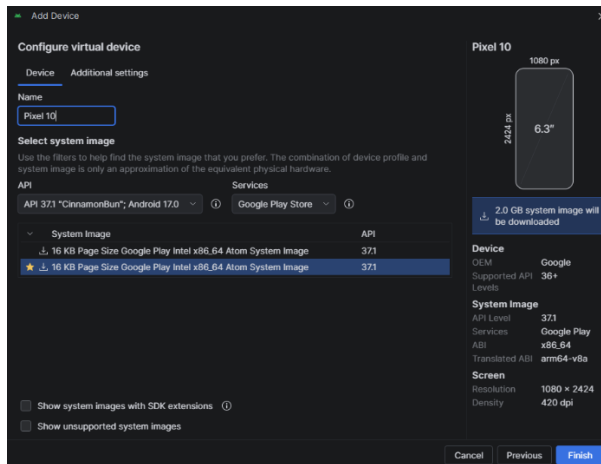
Now that Android Studio is installed and configured, you can create a virtual device to run your apps on. If you have previously installed Android Studio and configured emulators, they will appear here. If not, you may have a default device installed. To create a new emulator, click the “+” in the menu bar.



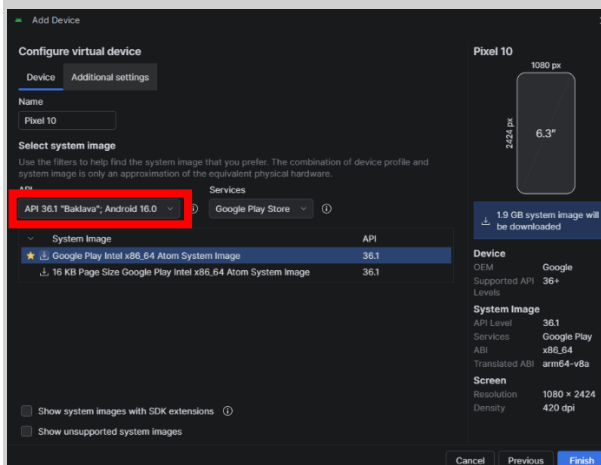
Choose the device you would like to emulate. Select form factor “Phone” and then the desired device from the list on the right (Pixel 10 is selected for this demo), and then click “Next”.



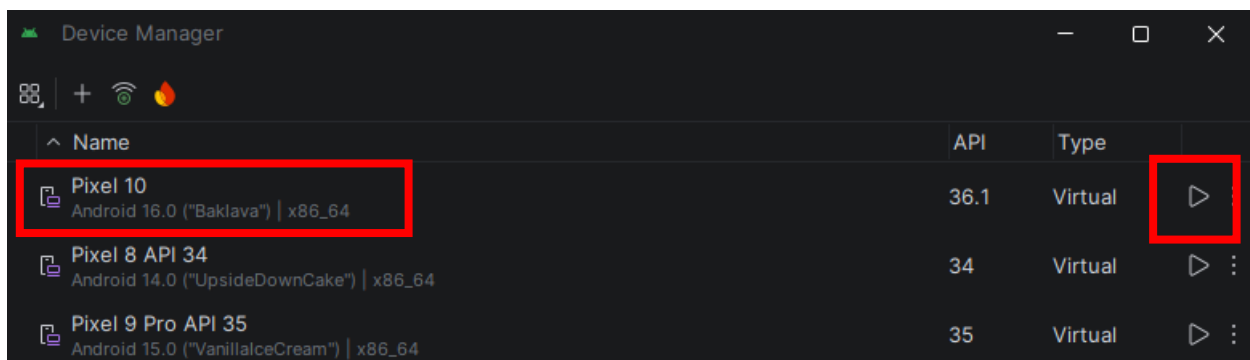
On the next screen, you can name your device and update any of the default settings. For this exercise, accept the default settings. If you experience issues running your emulator in the future, you can always edit these settings to adapt the device for your machine. Click “Finish”,



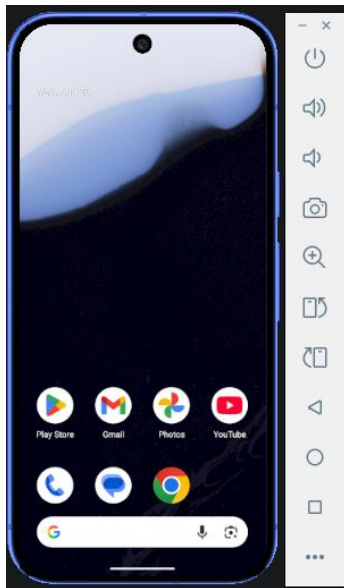
NOTE: at the time of this writing, Platform 17 was unavailable to download (due to 404 error), so Platform 16 was used in its place.



Once completed, you should see your new device listed and be able to run it by clicking the Play button:



If everything has gone well, you should see the emulated device starting up (it may take some time for the first startup, so please be patient – I waited almost 5 minutes on my machine).



Okay, that's it for now. In the future, you won't need to run Android Studio to launch the emulator. Your instructor will demonstrate how to launch it directly from VS Code. You can now safely close Android Studio, the emulator will continue to run.

Java Setup

To successfully build and run Android apps, you need to have the JDK installed and accessible.

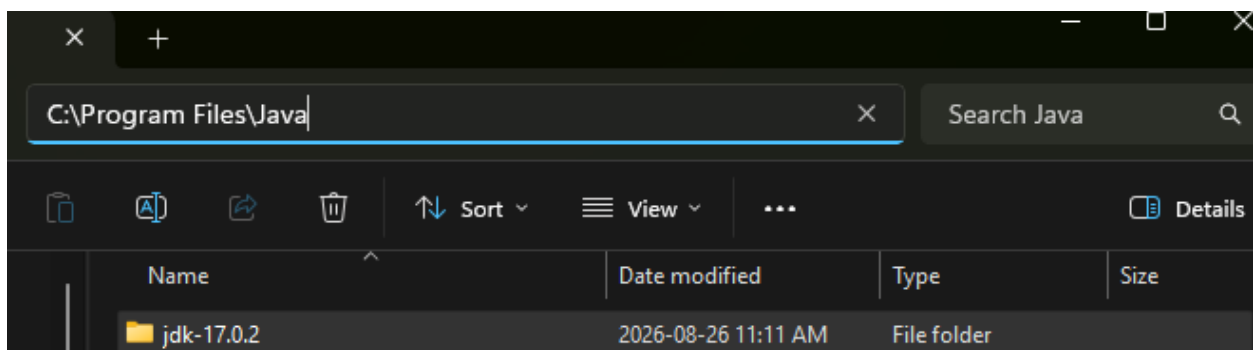
Download and install Java 17 (later versions will not work with the current version of Expo):

[Archived OpenJDK GA Releases](#)

NOTE: you can also download Java 17 from Oracle, but you will need to create an Oracle account to do so.

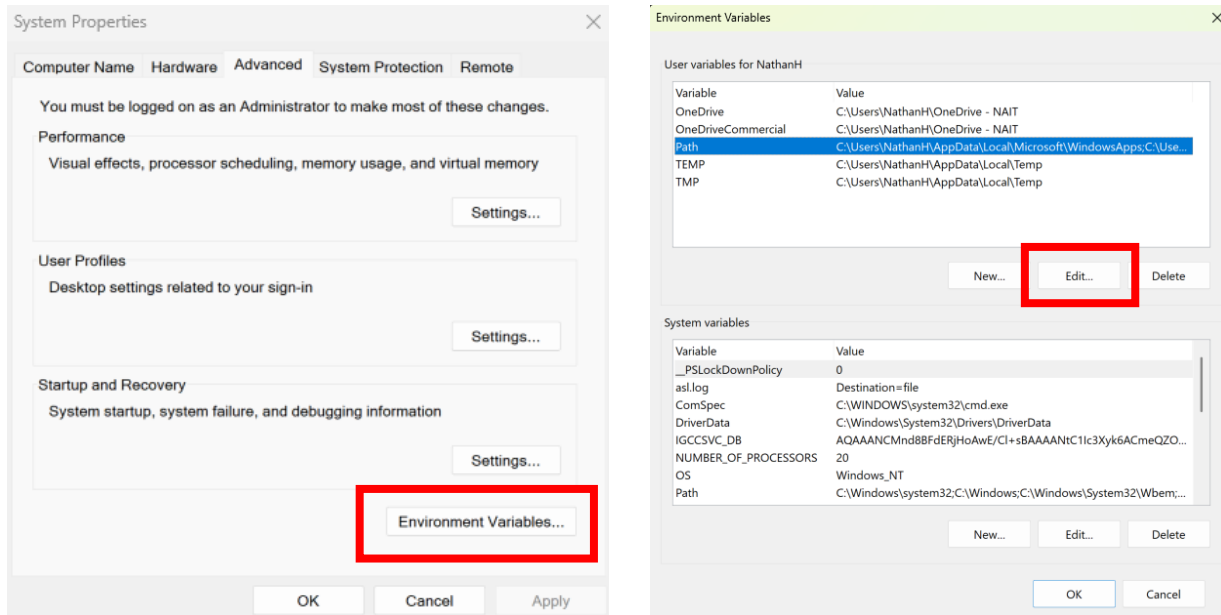
From the download page, select 17.0.2 (build 17.0.2+8) Windows download. A zip will be downloaded to your machine. Create a directory for the Java JDK (recommend C:\Program Files\Java\jdk-17.02\jdk-17.0.2) and extract the contents of the zip into this directory. The final location of the install will be:

C:\Program Files\Java\jdk-17.02\

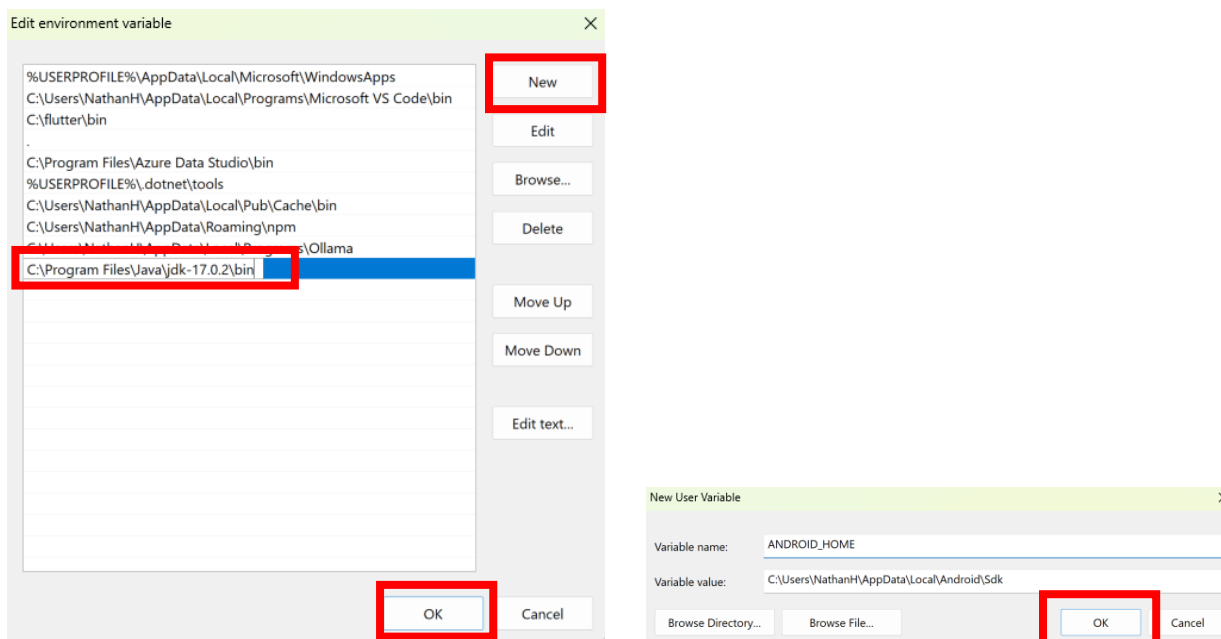


You will now need to add the java and javac executables to your PATH environment variable. This can be done by adding the C:\Program Files\Java\jdk-17.02\bin\ directory to your path.

From your Windows menu, type in “environment” and Select the “Edit the system environment variables” match. This will bring up the System Properties window. Click “Environment Variables...” and then choose to “Edit...” the user variable Path.



Create a “New” entry for the Java install. Once done, click “Ok to return to the Environment Variables window. At this time, also add the required ANDROID_HOME environment variable. Click “New”, under User variables, and then enter the new variable. Enter the location of the SDK you recorded earlier and click “Ok”.



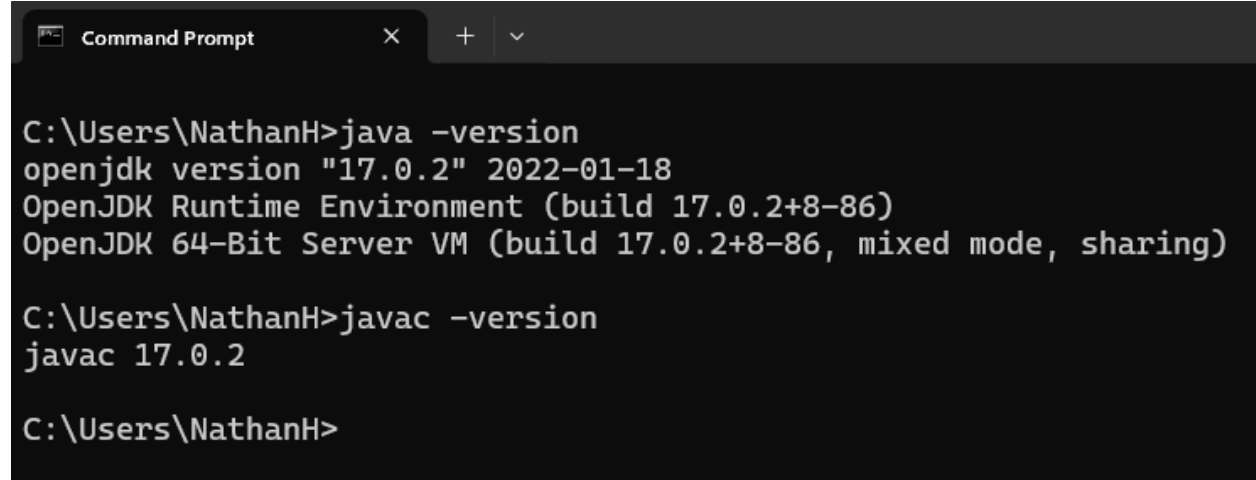
Now, click “Ok” again on the Environment Variables window, and then once more on the System Properties window.

To test the install, open a new cmd or ps window and run the following commands:

```
java -version
```

```
javac -version
```

You should see messages similar to the following:

A screenshot of a Windows Command Prompt window. The title bar says 'Command Prompt'. The command prompt shows the user's location as 'C:\Users\NathanH'. The user has entered 'java -version' and the output is: 'openjdk version "17.0.2" 2022-01-18', 'OpenJDK Runtime Environment (build 17.0.2+8-86)', and 'OpenJDK 64-Bit Server VM (build 17.0.2+8-86, mixed mode, sharing)'. The user then enters 'javac -version' and the output is 'javac 17.0.2'. The prompt returns to 'C:\Users\NathanH>'.

```
C:\Users\NathanH>java -version
openjdk version "17.0.2" 2022-01-18
OpenJDK Runtime Environment (build 17.0.2+8-86)
OpenJDK 64-Bit Server VM (build 17.0.2+8-86, mixed mode, sharing)

C:\Users\NathanH>javac -version
javac 17.0.2

C:\Users\NathanH>
```

Build and Run Your App

From the terminal in your projects directory, run the following two commands:

```
npx create-expo-app@latest test-app --template blank@sdk-57
```

```
cd test-app
```

```
npx expo install expo-dev-client
```

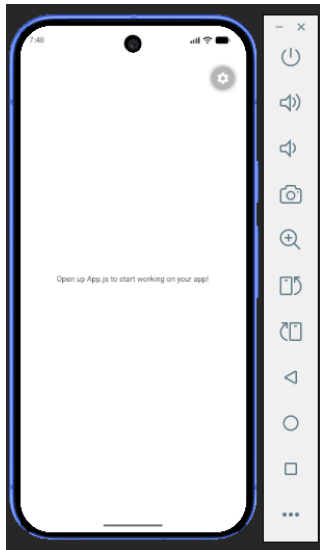
This will install a starter React Native project in your directory. To run the app, run the following command:

```
npx expo run:android
```

If everything goes well, you will see the default application built and running on your emulator.

NOTE: the first build will take some time, have patience.

At this point, you should have a successful Android/React Native development environment setup. You should see the default app running in the emulator.



Install VS Code Extensions

Install Expo Tools for VS Code to get a smoother React Native developer experience.

[Expo Tools - Visual Studio Marketplace](#)